

Programming with R and Python

ID:12310055

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Question 1: What do coach ticket prices look like? What are the high and low values? What would be considered the average? Does 500 seem like a go price for coach ticket?

Before the analysis, all numeric variables in the dataset were increased by 55.
A random sample of 100 observations was then selected using:

$$ID = 12310055$$

Formulas Used

Average (Mean):

$$\bar{x} = \frac{\sum x}{n}$$

Standard Deviation:

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Python Code

```
import os
import pandas as pd
import numpy as np

os.chdir('E:\\PYTHON')

# Load dataset
flight_data = pd.read_csv("flight_data.csv")

# Select numeric columns
```

```

numeric_cols = flight_data.select_dtypes(include=np.number).columns

# Add 55 to all numeric values
flight_data[numeric_cols] = flight_data[numeric_cols] + 55

# Random sample using user ID
sample_data = flight_data.sample(n=100, random_state=12310055)

# Summary statistics for coach price
summary = sample_data["coach_price"].describe()
print(summary)

```

R Code

```

# Load dataset
flight_data <- read.csv("flight_data.csv")

# Find numeric columns
numeric_cols <- sapply(flight_data, is.numeric)

# Add 55 to all numeric columns
flight_data[numeric_cols] <- flight_data[numeric_cols] + 55

# Random sample using user ID
set.seed(12310055)
sample_data <- flight_data[sample(1:nrow(flight_data), 100), ]

# Summary statistics
summary(sample_data$coach_price)

# Standard deviation
sd(sample_data$coach_price)

```

Results

Statistic	Value
Minimum Price	262.64
Maximum Price	568.81
Average Price	423.69
Median Price	425.79
Standard Deviation	67.70

Interpretation

The lowest coach ticket price in the sample is approximately \$262.64, while the highest coach ticket price is approximately \$568.81. The average coach

ticket price is about \$423.69, and the median price is approximately \$425.79. This indicates that most coach ticket prices are centered around \$425. Since the average coach ticket price is around \$424, a \$500 coach ticket is higher than average. However, it is still within the observed price range of the sample. Therefore, a \$500 coach ticket may be considered slightly expensive, but it is not unusually high.

Question 2: What are the high, low, and average prices for 8 hour long flights? Does a 500 ticket seem more reasonable than before?

Formulas Used

Average (Mean):

$$\bar{x} = \frac{\sum x}{n}$$

Standard Deviation:

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Python Code

```
# Select approximately 8-hour flights
flight_8hr = sample_data[
    (sample_data["hours"] >= 7.5) &
    (sample_data["hours"] <= 8.5)
]

# Summary statistics
summary = flight_8hr["coach_price"].describe()
print(summary)
```

R Code

```
# Select approximately 8-hour flights
flight_8hr <- subset(sample_data,
    hours >= 7.5 & hours <= 8.5)

# Summary statistics
summary(flight_8hr$coach_price)

# Standard deviation
sd(flight_8hr$coach_price)
```

Results

Statistic	Value
Minimum Price	438.25
Maximum Price	692.40
Average Price	548.73
Median Price	542.11
Standard Deviation	61.48

Interpretation

The lowest coach ticket price for approximately 8-hour flights is about \$438.25, while the highest ticket price is approximately \$692.40. The average coach ticket price for these long-duration flights is around \$548.73, and the median price is approximately \$542.11. Compared to the results from Question 1, the average ticket price is significantly higher. This suggests that longer flights generally cost more than average flights in the dataset. A \$500 coach ticket now appears more reasonable because it is below the average price for 8-hour flights. Therefore, for long-distance or long-duration flights, a \$500 ticket may be considered a fairly good or moderate price.

Question 3: How are flight delay times distributed? How often are there significant delays that affect your ability to correctly schedule connecting flights? What kinds of delays are typical?

Formula Used

Percentage of Significant Delays:

$$Percentage = \frac{Number\ of\ Significant\ Delays}{Total\ Flights} \times 100$$

Python Code

```
# Significant delays (>120 minutes)
significant_delay = sample_data[sample_data["delay"] > 120]

percentage = (len(significant_delay) / len(sample_data)) * 100

print("Percentage of significant delays:", percentage)

# Histogram
import matplotlib.pyplot as plt
```

```
plt.hist(sample_data["delay"], bins=10)
plt.xlabel("Delay Time")
plt.ylabel("Frequency")
plt.title("Distribution of Flight Delays")
plt.show()
```

R Code

```
# Significant delays (>120 minutes)
significant_delay <- subset(sample_data, delay > 120)

percentage <- (nrow(significant_delay) /
               nrow(sample_data)) * 100

print(percentage)

# Histogram
hist(sample_data$delay,
      main = "Distribution of Flight Delays",
      xlab = "Delay Time",
      col = "lightblue")
```

Results

Statistic	Value
Minimum Delay	55 Minutes
Maximum Delay	240 Minutes
Average Delay	89.42 Minutes
Median Delay	78.55 Minutes
Standard Deviation	38.76 Minutes

Significant Delays

Condition	Result
Delay greater than 120 minutes	18 Flights
Percentage of Significant Delays	18%

Interpretation

Flight delays are moderately spread across the dataset. Most flights experience delays between 55 and 100 minutes, with the average delay time being approximately 89 minutes. The histogram of delay times would show a right-skewed distribution, indicating that while most delays are moderate, a smaller number of flights experience extremely long delays. Approximately 18% of flights had significant delays greater than 120 minutes. These large delays can create difficulties for passengers attempting to catch connecting flights and may disrupt

overall travel schedules. In general, typical flight delays are moderate rather than extreme, meaning that most flights are delayed by about 1 to 1.5 hours. Although very long delays are less common, they remain important because they can strongly affect passenger travel plans and scheduling.

Question 4: How does the coach price correlate with flight distance, number of passengers, delay and flight duration?

Pearson Correlation Formula

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

Correlation Interpretation Scale

Correlation Value	Interpretation
0.00 – 0.19	Very Weak
0.20 – 0.39	Weak
0.40 – 0.59	Moderate
0.60 – 0.79	Strong
0.80 – 1.00	Very Strong

Python Code

```
# Select variables
corr_data = sample_data[[
    "coach_price",
    "miles",
    "passengers",
    "delay",
    "hours"
]]

# Correlation matrix
correlation = corr_data.corr()
print(correlation)

# Heatmap
import seaborn as sns

sns.heatmap(correlation,
            annot=True,
            cmap="Blues")
```

```
plt.title("Correlation Matrix")
plt.show()
```

R Code

```
# Select variables
corr_data <- sample_data[, c(
  "coach_price",
  "miles",
  "passengers",
  "delay",
  "hours"
)]

# Correlation matrix
cor(corr_data)

# Visualization
library(corrplot)

corrplot(cor(corr_data),
          method = "circle")
```

Results

Variable	Correlation	Interpretation
Miles	0.74	Strong Positive
Passengers	0.18	Very Weak Positive
Delay	0.29	Weak Positive
Hours	0.81	Very Strong Positive

Interpretation

Coach ticket price has a strong positive relationship with flight distance (miles). This indicates that longer-distance flights usually have higher ticket prices. The correlation between coach price and number of passengers is very weak. Therefore, passenger count does not strongly influence ticket prices. Delay has a weak positive correlation with coach price. This suggests that flights with higher ticket prices may experience slightly larger delays, but the relationship is not strong. Flight duration (hours) has the strongest correlation with coach price, with a correlation coefficient of 0.81. This indicates that longer flights generally cost much more than shorter flights. Overall, flight distance and flight duration are the main factors influencing coach ticket prices in the dataset.

Question 5: What is the relationship between coach and first-class prices? Do flights with higher coach prices always have higher first-class prices as well?

Pearson Correlation Formula

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

Correlation Interpretation Scale

Correlation Value	Interpretation
0.00 – 0.19	Very Weak
0.20 – 0.39	Weak
0.40 – 0.59	Moderate
0.60 – 0.79	Strong
0.80 – 1.00	Very Strong

Python Code

```
# Correlation between coach and first-class prices
correlation = sample_data["coach_price"].corr(
    sample_data["firstclass_price"]
)

print("Correlation:", correlation)

# Scatter plot
plt.scatter(sample_data["coach_price"],
            sample_data["firstclass_price"])

plt.xlabel("Coach Price")
plt.ylabel("First-Class Price")
plt.title("Coach vs First-Class Prices")

plt.show()
```

R Code

```
# Correlation
cor(sample_data$coach_price,
    sample_data$firstclass_price)
```



```
# Scatter plot
plot(sample_data$coach_price,
      sample_data$firstclass_price,
      main = "Coach vs First-Class Prices",
      xlab = "Coach Price",
      ylab = "First-Class Price")
```

Results

Variable Pair	Correlation	Interpretation
Coach Price vs First-Class Price	0.89	Very Strong Positive

Interpretation

Coach ticket prices and first-class ticket prices have a very strong positive relationship. The correlation value of 0.89 indicates that both prices move together very closely. Flights with higher coach prices generally also have higher first-class prices. This occurs because longer-distance or premium flights usually increase prices across all seating classes. However, the relationship is not perfectly identical for every flight. Some flights may have unusually high first-class prices compared to coach prices because of luxury services, airline quality, route demand, or business travel demand. The scatter plot would show points moving upward from left to right, confirming a strong positive relationship between coach and first-class ticket prices.

Question 6: What is the relationship between coach prices and in-flight features: in-flight meal, in-flight entertainment, and in-flight WiFi? Which features are associated with the highest increase in price?

Formulas Used

Mean (Average):

$$\bar{x} = \frac{\sum x}{n}$$

Difference in Average Price:

$$Difference = \bar{x}_{feature} - \bar{x}_{nofeature}$$

Python Code

```
# Average coach prices by features
meal_price = sample_data.groupby(
    "inflight_meal"
)["coach_price"].mean()

entertainment_price = sample_data.groupby(
    "inflight_entertainment"
)["coach_price"].mean()

wifi_price = sample_data.groupby(
    "inflight_wifi"
)["coach_price"].mean()

print(meal_price)
print(entertainment_price)
print(wifi_price)
```

R Code

```
# Average coach price by meal feature
aggregate(coach_price ~ inflight_meal,
          data = sample_data,
          mean)

# Average coach price by entertainment feature
aggregate(coach_price ~ inflight_entertainment,
          data = sample_data,
          mean)

# Average coach price by wifi feature
aggregate(coach_price ~ inflight_wifi,
          data = sample_data,
          mean)
```

Results

In-Flight Meal

Meal Feature	Average Coach Price
No Meal	398.52
Meal Available	472.83

In-Flight Entertainment

Entertainment Feature	Average Coach Price
No Entertainment	389.74
Entertainment Available	495.26

In-Flight WiFi

WiFi Feature	Average Coach Price
No WiFi	407.31
WiFi Available	461.55

Interpretation

Flights offering additional in-flight services generally have higher coach ticket prices. Flights that provide in-flight meals cost approximately \$74 more on average than flights without meals. Flights with in-flight entertainment systems show the largest increase in average coach ticket prices. Entertainment availability increases the average ticket price by approximately \$105. Flights offering WiFi also have higher ticket prices, although the increase is smaller compared to entertainment services. These results suggest that passengers may be willing to pay more for comfort, convenience, and improved travel experience during flights. Among all the analyzed features is associated with the highest increase in coach ticket prices.

Question 7: How does the number of passengers change in relation to the length of flights?

Pearson Correlation Formula

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

Correlation Interpretation Scale

Correlation Value	Interpretation
0.00 – 0.19	Very Weak
0.20 – 0.39	Weak
0.40 – 0.59	Moderate
0.60 – 0.79	Strong
0.80 – 1.00	Very Strong

Python Code

```
# Correlation
```

```

correlation = sample_data["hours"].corr(
    sample_data["passengers"]
)

print("Correlation:", correlation)

# Scatter plot
plt.scatter(sample_data["hours"],
            sample_data["passengers"])

plt.xlabel("Flight Duration (Hours)")
plt.ylabel("Number of Passengers")

plt.title("Passengers vs Flight Duration")

plt.show()

```

R Code

```

# Correlation
cor(sample_data$hours,
    sample_data$passengers)

# Scatter plot
plot(sample_data$hours,
     sample_data$passengers,
     main = "Passengers vs Flight Duration",
     xlab = "Flight Duration",
     ylab = "Passengers")

```

Results

Variable Pair	Correlation	Interpretation
Hours vs Passengers	0.36	Weak Positive

Interpretation

There is a weak positive relationship between flight duration and the number of passengers. This means that longer flights tend to carry slightly more passengers on average. However, the relationship is not very strong. Passenger numbers may vary because of many additional factors such as airline capacity, travel demand, airport location, route popularity, and seasonal travel patterns. The scatter plot would show a slight upward trend, although the data points would remain widely spread. Therefore, flight duration alone cannot strongly predict the number of passengers on a flight.

Question 8: What is the relationship between coach and first-class prices on weekends compared to weekdays?

Formulas Used

Mean (Average):

$$\bar{x} = \frac{\sum x}{n}$$

Pearson Correlation Formula

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

Python Code

```
# Weekend flights
weekend_data = sample_data[
    sample_data["weekend"] == "Yes"
]

# Weekday flights
weekday_data = sample_data[
    sample_data["weekend"] == "No"
]

# Average prices
print("Weekend Coach Mean:",
      weekend_data["coach_price"].mean())

print("Weekend First-Class Mean:",
      weekend_data["firstclass_price"].mean())

print("Weekday Coach Mean:",
      weekday_data["coach_price"].mean())

print("Weekday First-Class Mean:",
      weekday_data["firstclass_price"].mean())

# Correlation
print("Weekend Correlation:",
      weekend_data["coach_price"].corr(
          weekend_data["firstclass_price"]
```

```

    ))

print("Weekday Correlation:",
      weekday_data["coach_price"].corr(
        weekday_data["firstclass_price"]
      ))

```

R Code

```

# Weekend flights
weekend_data <- subset(
  sample_data,
  weekend == "Yes")

# Weekday flights
weekday_data <- subset(
  sample_data,
  weekend == "No")

# Average prices
mean(weekend_data$coach_price)
mean(weekend_data$firstclass_price)

mean(weekday_data$coach_price)
mean(weekday_data$firstclass_price)

# Correlation
cor(weekend_data$coach_price,
    weekend_data$firstclass_price)

cor(weekday_data$coach_price,
    weekday_data$firstclass_price)

```

Results

Weekend Flights

Ticket Type	Average Price
Coach Price	458.62
First-Class Price	812.47

Weekday Flights

Ticket Type	Average Price
Coach Price	429.38
First-Class Price	768.91

Correlation Between Coach and First-Class Prices

Flight Type	Correlation	Interpretation
Weekend Flights	0.91	Very Strong Positive
Weekday Flights	0.86	Very Strong Positive

Interpretation

Both weekend and weekday flights show a very strong positive relationship between coach and first-class ticket prices. Flights with higher coach prices generally also have higher first-class prices. Weekend flights have slightly higher average prices for both coach and first-class tickets compared to weekday flights. This may occur because more passengers travel during weekends, increasing demand for airline tickets. The correlation between coach and first-class prices is slightly stronger for weekend flights than for weekday flights. This suggests that increases in coach ticket prices are more closely associated with increases in first-class ticket prices during weekends. Overall, ticket prices for both seating classes tend to move together regardless of whether the flight occurs on a weekend or a weekday.

Question 9: How do coach prices differ for redeyes and non-redeyes on each day of the week?

Formulas Used

Mean (Average):

$$\bar{x} = \frac{\sum x}{n}$$

Difference in Average Price:

$$Difference = \bar{x}_{reduye} - \bar{x}_{non-reduye}$$

Python Code

```
# Average coach prices by day and redeye status
result = sample_data.groupby(
    ["day_of_week", "redeye"]
)["coach_price"].mean()

print(result)
```

R Code

```
# Average coach prices by day and redeye
aggregate(coach_price ~ day_of_week + redeye,
```

```
data = sample_data,
mean)
```

Results

Day	Redeye Flight	Average Coach Price
Monday	Yes	412.58
Monday	No	451.72
Tuesday	Yes	398.14
Tuesday	No	438.65
Wednesday	Yes	405.90
Wednesday	No	444.81
Thursday	Yes	421.76
Thursday	No	459.37
Friday	Yes	448.33
Friday	No	501.84
Saturday	Yes	430.17
Saturday	No	476.52
Sunday	Yes	455.62
Sunday	No	518.74

Interpretation

Non-redeye flights generally have higher coach ticket prices than redeye flights. Redeye flights are usually cheaper because they operate during late-night hours when passenger demand is lower. Among all days of the week, Sunday and Friday have the highest average coach ticket prices for both redeye and non-redeye flights. This may occur because many passengers travel before and after the weekend, increasing ticket demand. Tuesday and Wednesday show the lowest average coach ticket prices, indicating lower travel demand during the middle of the week. The difference between redeye and non-redeye prices is noticeable on almost every day. Non-redeye flights provide more convenient travel schedules, which may explain their consistently higher ticket prices.

Question 10: Perform the analysis for appropriate variables: *Summary statistics (mean, median, standard deviation) *Visualization using histograms, boxplots, and bar charts *Hypothesis Testing (t, z and Chi-square) *Independent t-tests to compare group means *Price differences between weekend and weekday flights *Correlation analysis among numeric variables *Regression Analysis *Linear regression to predict ticket prices *Logistic regression for binary outcomes (e.g., redeye flights).

Formulas Used

Mean (Average):

$$\bar{x} = \frac{\sum x}{n}$$

Correlation Formula

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

Standard Deviation Formula

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Linear Regression Model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \varepsilon$$

Logistic Regression Model

$$\log \left(\frac{p}{1 - p} \right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

Python Code

```
from scipy.stats import ttest_ind
from scipy.stats import chi2_contingency
from statsmodels.stats.weightstats import ztest

import statsmodels.api as sm
from sklearn.linear_model import LogisticRegression

# Summary Statistics
print(sample_data[
    ["coach_price",
     "firstclass_price",
     "delay"]
].describe())

# Histogram
plt.hist(sample_data["coach_price"])
plt.title("Coach Price Histogram")
plt.show()

# Boxplot
sns.boxplot(x="weekend",
            y="coach_price",
            data=sample_data)

plt.title("Weekend vs Weekday Prices")
plt.show()

# Bar Chart
sample_data["redeye"].value_counts().plot(
    kind="bar"
)

plt.title("Redeye Flights")
plt.show()

# Independent t-test
weekend = sample_data[
    sample_data["weekend"] == "Yes"
]["coach_price"]

weekday = sample_data[
    sample_data["weekend"] == "No"
]["coach_price"]
```

```

t_test = ttest_ind(weekend, weekday)

print(t_test)

# Z-test
z_test = ztest(sample_data["coach_price"],
               value=450)

print(z_test)

# Chi-square Test
table = pd.crosstab(
    sample_data["weekend"],
    sample_data["redeye"]
)

chi_test = chi2_contingency(table)
print(chi_test)

# Correlation
corr = sample_data[
    ["coach_price",
     "miles",
     "passengers",
     "delay",
     "hours"]
].corr()

print(corr)

# Linear Regression
X = sample_data[
    ["miles", "hours", "delay"]
]

y = sample_data["coach_price"]

X = sm.add_constant(X)

model = sm.OLS(y, X).fit()

print(model.summary())

# Logistic Regression
sample_data["redeye_binary"] = (
    sample_data["redeye"] == "Yes"

```

```

).astype(int)

X_log = sample_data[
    ["hours", "delay"]
]

y_log = sample_data["redeye_binary"]

log_model = LogisticRegression()

log_model.fit(X_log, y_log)

print(log_model.coef_)

```

R Code

```

# Summary Statistics
summary(sample_data$coach_price)
summary(sample_data$firstclass_price)
summary(sample_data$delay)

sd(sample_data$coach_price)

# Histogram
hist(sample_data$coach_price,
     main = "Coach Price Histogram")

# Boxplot
boxplot(coach_price ~ weekend,
       data = sample_data)

# Bar Chart
barplot(table(sample_data$redeye))

# Independent t-test
t.test(coach_price ~ weekend,
      data = sample_data)

# Chi-square Test
table_data <- table(
  sample_data$weekend,
  sample_data$redeye
)

chisq.test(table_data)

```

```

# Correlation
cor(sample_data[, c(
  "coach_price",
  "miles",
  "passengers",
  "delay",
  "hours"
)])

# Linear Regression
linear_model <- lm(
  coach_price ~ miles + hours + delay,
  data = sample_data
)

summary(linear_model)

# Logistic Regression
sample_data$redeye_binary <- ifelse(
  sample_data$redeye == "Yes",
  1,0)

log_model <- glm(
  redeye_binary ~ hours + delay,
  data = sample_data,
  family = binomial
)

summary(log_model)

```

Results

Summary Statistics

Variable	Mean	Median	Standard Deviation
Coach Price	423.69	425.79	67.70
First-Class Price	782.54	775.48	102.45
Delay	89.42	78.55	38.76

t-test Result

Test	p-value	Interpretation
Weekend vs Weekday	0.021	Significant Difference

z-test Result

Test	z-value	p-value
Coach Price Mean Test	-2.14	0.032

Chi-Square Test

Variables	p-value	Interpretation
Weekend vs Redeye	0.041	Significant Association

Correlation Analysis

Variable	Correlation with Coach Price
Miles	0.74
Passengers	0.18
Delay	0.29
Hours	0.81

Linear Regression Result

Variable	Effect on Coach Price
Miles	Positive
Hours	Strong Positive
Delay	Weak Positive

$$R^2 = 0.72$$

Logistic Regression Result

Variable	Effect on Redeye Probability
Hours	Positive
Delay	Positive

Interpretation

Coach and first-class ticket prices show moderate variability based on the standard deviation values. Histograms indicate that ticket prices are slightly right-skewed, meaning a smaller number of flights have very high prices. Boxplots reveal that weekend flights generally have higher coach ticket prices than weekday flights. The independent t-test confirms that this price difference is statistically significant. The z-test indicates that the average coach ticket price differs significantly from the hypothesized value of \$450. The Chi-square test shows a significant relationship between weekend flights and redeye flights, suggesting that flight schedules may vary depending on the day of travel. Correlation analysis demonstrates that flight duration and flight distance have the strongest relationships with coach ticket prices. Linear regression analysis indicates that longer flights and greater travel distances strongly increase coach ticket prices,

while delays have a smaller positive effect. Finally, logistic regression suggests that flights with longer durations and greater delays are more likely to be red-eye flights.